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Comprehensive gridded dataset of photosynthetically active radiation in the upper ocean from 1958 to 2022

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ABSTRACT

Photosynthetically Active Radiation (PAR) plays a crucial role in shaping marine ecosystems, influencing primary production, species interaction, and phytoplankton seasonal dynamics. However, comprehensive long-term (gap-free) datasets for both surface PAR and the diffuse Attenuation Coefficient of Photosynthetically Active Radiation (K_dPAR) are currently lacking. In this study, we introduce two new extensive global 4D PAR gap-free datasets (Longitude x Latitude x Day x Depth) at a resolution of 0.25° latitude x 0.25° longitude from surface to 250 m covering the periods 1998–2022 and 1958–2022. The first dataset (1998–2022) is primarily derived from Globcolour (surface PAR and Chlorophyll-a), supplemented with missing surface PAR data estimated using the Environmental String Model (ESM) with key climatic ERA5 variables. Missing Chlorophyll-a data are interpolated by applying the DINEOF method (Data Interpolating Empirical Orthogonal Functions) and transformed into K_dPAR. Visual and numerical evaluations closely approximate observations, demonstrating the accuracy of our approach. Subsequently, we extend our dataset back to 1958 using exclusively the ESM based on key climatic ERA5 variables. The ESM outperforms the Generalized Regression on Neural Network (GRNN) in computational efficiency while yielding similar results. Validation against *in-situ* measurements confirms the reliability of PAR and K_dPAR surface products. Although the 1958–2022 dataset exhibits limited daily variability in PAR compared to the 1998–2022 dataset, it effectively captures critical spatial-temporal patterns, as demonstrated by correlative and comparative studies with El Niño indices. Furthermore, the similarity observed between euphotic depth (Z_{eu}) derived from our ESM-based 4D PAR dataset (1958–2022), and the Mercator-Ocean hindcast model, along with *in-situ* data, underscores the robustness of our approach in capturing light availability at depth.

1. Introduction

1.1. Description of PAR and its importance for marine sciences

Despite its microscopic size, phytoplankton is responsible for nearly 50% of Earth's primary production (NRC, 2000). Changes in their abundance and biodiversity have significant influences on the structure and function of marine ecosystems (Kléparski et al., 2021). Phytoplankton distribution is controlled by many biotic and abiotic factors (Banse, 1994; Geider et al., 2001). Among variables, light availability plays an important role because it influences photosynthesis and autotrophic carbon fixation, an influence that nevertheless depends upon species (Yang et al., 2020). Limited light availability can diminish or even inhibit photosynthesis, delay the onset of blooms, and cause competition among species for light (Huisman et al., 1999; Lowry et al., 2014; Lewis et al., 2019). Because of their greater surface-to-volume ratio that confers them a competitive advantage in waters with limited resources, smaller phytoplankton are able to utilize light more effectively than larger species (Burson et al., 2018; Edwards et al., 2011; Grover, 1989; Kiørboe, 1993). In contrast, an excess of light may reduce photosynthetic capacity, a process called photoinhibition (Platt et al., 1980).

Studying Photosynthetically Active Radiation (PAR), which is the range of wavelengths in the visible spectrum (*i.e.*, 400–700 nm) used by photosynthetic organisms (McCree, 1971), is crucial to understand large-scale horizontal and vertical phytoplankton distribution. PAR helps determine the euphotic zone (Z_{eu}), which extends to a depth where there is enough light for photosynthesis to occur and is often defined as the depth where PAR reaches 1% of its surface value (Parsons et al., 1984; Ryther, 1956; Smith and Kemp, 2001; Walsh et al., 1995).

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Moreover, PAR spatial and temporal changes are important because they affect regionally not only phytoplankton populations but also the phenology of some taxonomic groups such as crustaceans and fish (Arnone et al., 1998; Caracciolo et al., 2021; Kléparski et al., 2022).

1.2. Availability of PAR at surface and PAR at depth

Although implemented long after some biological sampling programs, remote sensing appears to be an effective way to improve our knowledge of the environmental conditions controlling phytoplankton distribution (Chassot et al., 2011). Among the many parameters computed, surface PAR is available since September 1997 and the beginning of the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) operational phase. However, the amount of daily surface PAR that reaches the Earth's surface measured by satellite sensors is affected by the atmosphere, particularly by aerosols, cloud cover, ozone, water vapor, and various gases (Runnström et al., 2006; Sarkar et al., 2006), whilst the ability to process data in high latitude regions is restricted because of high sensor zenith angle complications for atmospheric correction, or simply because of the curvature of the planet itself (Li et al., 2017; He et al., 2018; Ruddick et al., 2014). Latest sensors such as MODerate resolution Imaging Spectroradiometer-Aqua (MODIS-A) and Visible Infrared Imaging Radiometer Suite (VIIRS) mitigate these problems in the assessment of daily surface PAR while PAR algorithms can provide estimations under cloudy conditions (Frouin et al., 2003; Frouin et al., 2012). Nevertheless, these problems persist when it comes to the estimation of Diffuse Attenuation Coefficient of Photosynthetically Active Radiation (K_dPAR). Historically related to chlorophyll-a (Chla) concentration in clear open ocean waters (Morel, 1988), where phytoplankton is the main contributor to attenuation (Claustre and Maritorena, 2003), K_dPAR helps determine how well surface PAR penetrates into the ocean to compute Z_{eu}. As there is no daily estimation of K_dPAR due to cloud cover or errors in atmospheric corrections (ACRI-ST, 2007), spatial interpolation is necessary to complete these datasets. A full (*i.e.*, no missing data) 4D (Latitude × Longitude × Depth × Day) PAR dataset, especially in high latitudes, is critical to model 4D plankton distribution. Such a dataset could be associated to the Continuous Plankton Recorder (CPR), a plankton monitoring program that has sampled plankton since 1946 (Edwards et al., 2016). PAR determination across all latitudes would help in understanding the large-scale spatio-temporal distribution of plankton, their phenology and long-term shifts, as well as assessing the possibility for climate-induced alteration in depth distribution (Hutchinson, 1957; Beaugrand et al., 2015; Perry et al., 2005).

1.3. Objectives

While predictive modeling may be a good alternative because PAR can be estimated using radiative transfer models, both surface PAR and K_dPAR long-term datasets are not available yet (Frouin et al., 2018). Gattuso et al. (2006, 2020) used surface PAR and K_dPAR with Morel's eq. (2007) to estimate PAR in depth. However, these estimates are monthly-based and may not fully capture rapid phytoplankton dynamics, which can occur within minutes to hours, influencing photoinhibition and light harvesting (Dubinsky and Stambler, 2009; Lavaud, 2007; MacIntyre et al., 2000). The goal of this study is to propose two datasets. First, a complete remote sensing dataset for the period 1998–2022. Here missing PAR and K_dPAR data, especially prevalent at high latitudes in winter in the current Globcolour daily surface PAR and K_dPAR datasets were completed by means of the Environmental String Model (ESM hereafter, Beaugrand, 2024) and Data Interpolating Empirical Orthogonal Functions (DINEOF, Alvera-Azcárate et al., 2005). Second, these completed remote sensing datasets are subsequently used as inputs to estimate surface PAR and K_dPAR from 1958 to 2022 by using the ESM. Third, our method is compared with Generalized Regression on Neural Network (GRNN). Finally, a comprehensive assessment is conducted to evaluate the robustness of our product through statistical comparisons with other products (MERRA-2, Mercator-Ocean Hindcast model) and *in-situ* samples (SEABASS).

2. Data

2.1. Photosynthetically Active Radiation (PAR)

Level 3 daily mean just above the surface PAR (PAR_{0+}) estimated from SeaWiFS, MODIS-A, and VIIRS were downloaded from the NASA Ocean Biology Processing Group (OBPG) website (https://oceancolor.gsfc.nasa.gov) at a spatial resolution of 4 km from 1997 to 2022 and resampled on a 0.25° latitude × 0.25° longitude grid (~28 km at the equator). The different sensors have distinct periods of acquisition: SeaWiFS from September 1997 to December 2010, MODIS-A and VIIRS from May 2002 and January 2012 onwards, respectively. While the applied algorithms allow PAR estimation under cloudy condition (Frouin et al., 2003; Frouin et al., 2012), gaps still remain due to the orbit and sensing geometry and consequently a merging technique is necessary to reduce differences/biases among individual polar-orbiting sensors (Frouin et al., 2003; Frouin et al., 2012; Tan and Frouin, 2019).

The European Spatial Agency (ESA) undertook the GlobColour project (http://www.globcolour.info/) (ACRIST 2007) to deliver a merged and reliable satellite-derived data products on ocean color and related variable. The project employs the data-day method, extending a day's data from 24 to 28 h to ensure spatial consistency, with consideration for different satellite orbits and pixel validity based on sun zenith angle. However, it also acknowledges inter-mission inconsistencies in global time series aligned with changes in the availability of individual sensors (ACRI-ST, 2007). The level-2 daily remote sensing PAR_{0+} are produced for each sensor and merged as the weighted average of all common estimations. The spatial coverage remains uncomplete, however, because there still remains no data at high latitudes in the winter hemisphere, *i.e.*, for Sun Zenith Angle (SZA) > 70°, compared with individual products available in OBPG, while gaps due to the orbit and sensing geometry are reduced. The level 3 daily mean merged PAR_{0+} generated by Globcolour were downloaded directly from the website and resampled on a 0.25° latitude × 0.25° longitude grid. PAR data were then converted as follows:

$$PAR_{W.m^{-2}} = PAR_{E.m^{-2}d^{-1}} \times \frac{\beta}{\varepsilon \times \tau} \quad (1)$$

Where β is Avogadro's number (6.02×10^{23} quanta.E^{-1}), ε is the number of seconds per day (86,400 s.d^{-1}) and τ the factor to convert in the PAR range with an accuracy of a few percent (2.77×10^{18} quanta.s^{-1}.W^{-1}) (Morel and Smith, 1974; Kirk, 1994).

2.2. Chlorophyll-a concentration

To estimate K_dPAR, we downloaded the merged mapped global daily Chla "Cloud Free" level 4 product (1998–2022) from GlobColour (ACRI-ST, 2007) at the Copernicus Marine Environmental Monitoring Service (CMEMS) website (http://marine.copernicus.eu/services-portfolio/access-to-products/). To complete remote sensing missing data, additional modelled Chla (reanalysis) data were downloaded from the biogeochemical hindcast model provided by Mercator-Ocean (GLOBAL_MULTIYEAR_BGC_001_029) at a resolution of 1/12° and resampled on a 0.25° latitude × 0.25° longitude grid (CMEMS, 2020).

2.3. Environmental explanatory variables

Environmental explanatory data originated from the ERA5 database. Updated on a regular basis, it is a comprehensive dataset of atmospheric and meteorological observations and reanalysis products that is operated by the European Centre for Medium-Range Weather Forecasts (ECMWF) (Hersbach and Dee, 2016).The dataset covered the period

from 1958 to the present at an hourly time interval and a spatial resolution of 0.25° and was validated against observations from weather stations worldwide (Hersbach et al., 2020; Li et al., 2020; Lei et al., 2022). Many ERA5 variables were downloaded and daily averaged since 1958 directly on the Copernicus Climate datastore (https://cds.climate.copernicus.eu/): Mean Surface Downward Shortwave Radiation Flux Clear Sky (MSDCS), to have information on the amount of daily solar radiation that reaches a horizontal plane at the surface of the Earth during clear sky conditions; Surface Solar Radiation Downwards (SSRD) and Surface net Solar Radiation (SSR) to compute surface albedo in the solar spectral domain (A_s) following ERA5 report (Hogan, 2015):

$$A_s = 1 - \frac{SSR}{SSRD} \quad (2)$$

Total Cloud Liquid Water (TCL) and Medium Cloud Cover (MCC) were used for atmospheric conditions that mitigate solar transmission to the surface; Sea Surface Temperature (SST) was downloaded and linked to Chla estimations; Sea Ice Concentration (SICONC) was used to remove data located upon a threshold SICONC value of 0.1 for PAR_{0+} and Chla. Then, the Zonal and Meridional component of the surface Stokes drift (UST, VST) were utilized to calculate the Stokes drift (WST), which is the net drift velocity due to surface wind waves and approximately corresponds to 2% of the local wind speed (Ardhuin et al., 2009). WST daily anomalies were calculated by subtracting the average value of each geographical cell in the 25-year data, as wind anomalies may exhibit regional correlation patterns with Chla concentration (Kahru et al., 2010).

The minimum daily SZA, a factor not considered in MSDCS, was calculated daily by using the spherical trigonometric relationship between SZA, the latitude and the hour angle determined from the position of the shadow over time (Shivalingaswamy and Kagali, 2017).

Bathymetry data (BTM, GEBCO2021 Grid) was downloaded from the General Bathymetric Chart of the Oceans (GEBCO) website (https://www.gebco.net/data_and_products/gridded_bathymetry_data/#global). For the purpose of the analysis, these data were divided into four classes: 0–50 m, 50–100 m, 100–200 m and > 200 m. The choice of these classes was determined to identify zones where bathymetry constraints have significant influences on ocean physical properties (Bouffard et al., 2014; Escudier et al., 2013).

Distances from the nearest coastline map (DTC), generated by the OBPG website, was downloaded at a resolution of 0.01° and resampled on a 0.25° latitude × 0.25° longitude grid. For the purpose of the analysis, these data were divided into four classes: 0–25 km, 25–50 km, 50–100 km, and > 100 km. The choice of these classes was made to differentiate coastal distances from the shoreline, as surface Chla concentration falls off with increasing distance to land (*e.g.*, Johnson et al., 2007).

2.4. Variables and products used for evaluation

Remotely sensed daily surface PAR_{0+} originated from the Earth Polychromatic Imaging Camera (EPIC) onboard DSCOVR (Frouin et al., 2022) was downloaded from the NASA Langley ASDC website at a 0.17° resolution and resampled on a 0.25° latitude × 0.25° longitude grid. PAR_{0+} estimations originated from daily multiple EPIC observations during the day using a budget approach with NCEP Reanalysis 2, MERRA-2, and EPIC Level 2 data for atmospheric parameters, enabling the production of comprehensive PAR_{0+} data to compare with our ESM-based outputs (Hao et al., 2019; Hao et al., 2020).

Data on *in-situ* PAR_{0+}, Diffuse Attenuation Coefficient at 490 nm (K_d490) and euphotic depth (Z_{eu}, m) originated from the SeaWiFS Bio-optical Archive and Storage System (SEABASS) Validation Search, developed as part of the SeaWiFS Project (Werdell et al., 2002; Hooker et al., 1995), namely the NASA bio-Optical Marine Algorithm Dataset (NOMAD), was downloaded from the NASA Ocean Color website (https://seabass.gsfc.nasa.gov/search_results/val). Only PAR data from the CERES Ocean Validation Experiment (COVE) platform in the coastal Atlantic Ocean were used (Rutledge et al., 2006).

Modelled Surface Downwelling PAR_{0+} Diffuse Flux and Surface Downwelling PAR_{0+} Beam Flux from the MERRA-2 reanalysis database (Gelaro et al., 2017) were downloaded at 1-h intervals with spatial resolution of approximately 0.5°. Both PAR products were integrated, averaged daily and resampled on a 0.25° latitude × 0.25° longitude grid.

Data from BGC-Argo floats profiles were downloaded since 2016 using BGC-Argo toolbox for R (Cornec et al., 2021). Among all profiles downloaded that have a Chla fluorometer, only ascending profiles were kept. Only "Chla_adjusted" data were kept on ascending profiles and point with a QC of 1 (Good data), 2 (Probably good data), 5 (Value changed) and 8 (Estimated value). Chla was computed as the simple average of the BGC samples over the optical depth defined as $1/K_d490$ following Bretagnon and Mangin (2021). A total of 54,081 profiles was downloaded.

Three indices were downloaded to examine the relationships between PAR anomalies and El Niño: (i) The Southern Oscillation Index (SOI), (ii) the Multivariate El Niño Southern Oscillation (ENSO) Index (MEI) and (iii) El Niño Modoki Index (EMI). The SOI represents the difference between the normalized sea level pressure anomalies at Tahiti and Darwin (Ropelewski and Jones, 1987). The MEI represents the temporal sequence of the primary composite Empirical Orthogonal Function (EOF) derived from five distinct variables within the tropical Pacific basin (Wolter and Timlin, 1993). These first two indices were downloaded from the National Weather Service Climate Prediction Center (NWSCPC, 2023). For a comparison purpose, the MEI utilized by Tan and Frouin (2019) was downloaded here (https://psl.noaa.gov/enso/mei.old/table.html). Finally, the EMI was downloaded from the Japan Agency for Marine-Earth Science and Technology (JAMEST, Ashok et al., 2007).

Modelled Z_{eu} (m) were downloaded from the Global Ocean Low and Mid Trophic Levels Biomass Content Hindcast available in CMEMS (GLOBAL_MULTIYEAR_BGC_001_033). This model, which takes as input PAR from ERA-interim, was provided by Mercator-Ocean at a resolution of 1/12° and resampled on a 0.25° latitude × 0.25° longitude grid (CMEMS, 2022).

Downloaded data and their origin were summarized in Supplementary Table 9.

3. Methods

3.1. Algorithm development and evaluation plan

In this study, the objectives were to provide a full 4D (Latitude × Longitude × Depth × Day) gridded dataset of PAR at 0.25° latitude × 0.25° longitude resolution for the period 1958–2022. For the period where remote sensing data were available (1998–2022, Section 3.3), we first designed a numerical procedure based on the Environmental String Model (ESM) to fully assess PAR_{0+} (Section 3.3.1). ESM results were compared with another method based on artificial intelligence (*i.e.*, Generalized Regression based on Neural Network, GRNN, Section 3.3.1) and compared with EPIC DSCOVR PAR_{0+} estimation since 2015 and *in-situ* dataset from SEABASS (Section 3.3.2). Then, we estimated K_dPAR from Chla concentration, where missing data were estimated from Data Interpolating Empirical Orthogonal Functions (DINEOF, Section 3.3.3) and compared to BGC-Argo Chla data (Section 3.3.4). For 1958–2022 (Section 3.4), we applied the ESM to estimate PAR_{0+} and K_dPAR using the completed datasets for 1998–2022; here also we compared the ESM with GRNN (Section 3.4.1). Tests and bias corrections were carried out on surface PAR and K_dPAR products in two ways (Section 3.4.2): (i) comparison with *in-situ* dataset and modelled data from MERRA-2 and Mercator-Ocean hindcast models, (ii) EOF analyses on two periods following published studies to examine the influence of El Niño on monthly PAR_{0+}. PAR was computed just below the surface (PAR_{0-}) by means of a linear model based on MERRA-2 surface albedo data. Finally,

we completely estimated PAR at depth from both PAR_{0-} and K_dPAR using the Beer-Lambert law (Section 3.5).

3.2. Statistical metrics

Numerical comparisons were evaluated throughout the numerical procedure using four quantitative statistical metrics: (i) the Root Mean Square Deviation (RMSD), (ii) the Mean Bias (MB), (iii) the Mean Absolute Deviation (MAD), and (iv) the Pearson's correlation ($r_{Pearson}$):

$$RMSD = \sqrt{\frac{\sum_{g=1}^{b}\left(f_g - o_g\right)^2}{b}} \tag{3}$$

$$MAD = \frac{\sum_{g=1}^{b}\left|f_g - o_g\right|}{b} \tag{4}$$

$$MB = \frac{\sum_{g=1}^{b} f_g - o_g}{b} \tag{5}$$

$$r_{Pearson} = \frac{\sum_{g=1}^{b}\left(o_g - \bar{o}\right)\left(f_g - \bar{f}\right)}{\sqrt{\sum_{g=1}^{b}\left(o_g - \bar{o}\right)^2}\sqrt{\sum_{g=1}^{b}\left(f_g - \bar{f}\right)^2}} \tag{6}$$

Where f_g was the predicted value g, o_g the observed value g, $\bar{f}$ the mean of predicted values, $\bar{o}$ was the mean of observed values and b the number of pairs of points involved in the comparison. Pearson's correlation has then been squared as r^2, the coefficient of determination, to understand how well our model explained the variance. RMSD and MAD were also expressed as a percentage of the average observed values (*i.e.*, PRMSD, PMAD). As each of these metrics provides different perspective on model accuracy and performance, we made the choice to amalgamate the PMAD, PRMSD, and r^2 metrics into a simple Comprehensive Index (CI) to select the optimal parameters for our model and compare our results more effectively. This involved computing the mean of all three metrics, with the transformation of r^2 to 1-r^2, the lower CI indicating the best combination of parameters (Eq. 7). This metric was computed because neither PRMSD and PMAD exceeded 1 (or expressed as a percentage 100%).

$$CI = \frac{PMAD + PRMSD + (1 - r^2)}{3} \tag{7}$$

3.3. Period of remote sensing data availability (1998–2022, daily resolution)

3.3.1. Filing of missing data in the PAR Globcolour merged product

We first used an Ecological Niche Model (ENM) to fill gaps in the PAR_{0+} Globcolour merged product. We chose the Environmental String Model (ESM) because it is among the few (*e.g.*, Generalized Regression on Neural Network) that consider local relationships among variables (Beaugrand, 2024). The ESM conceptualizes the niche as an Environmental String (ES), which is defined as a combination of environmental variables which is associated to the value of a dependent (response) variable (here PAR_{0+} satellite observations). Modeling was performed in three steps:

Step 1. Environmental data were standardized between 0 and 1 to represent all ESs in a 2-dimensional Euclidean space. All environmental vectors were standardized as follows:

$$x^* = \frac{x - min(x)}{max(x) - min(x)} \tag{8}$$

Where x is the column vector of all values of a given parameter with the minimum and maximum preferably computed from large-scale environmental datasets (Beaugrand, 2024). This process ensures that the value of each environmental parameter falls between 0 (minimum value) and 1 (maximum value). This step was crucial as it allowed a visualization of the multi-dimensional nature of the ESs in a 2D space and an equal weighting of each variable on PAR_{0+} estimates. Each valid satellite PAR_{0+} observation, stored in A_n, was represented by an ES, stored in matrix C_{np}, where n is the number of samples and p is the total number of environmental variables.

Each missing satellite PAR_{0+} observation in the Globcolour merged product was associated to a matrix U_{mp}, with m being the number of observations where an estimation was necessary. This matrix contained all Target Environmental Strings (TESs), *i.e.*, combinations of environmental variables used to estimate PAR_{0+} values.

To ensure temporal and local relationships, ES and TES were computed for each day of the dataset and composed of the following environmental variables: SZA, MSDCS, MCC, and TCL. To increase the number of ESs in polar regions, we fixed PAR_{0+} to 0 when MSDCS = 0. Finally, available NASA OBPG individual sensor data were also added to the ESs. Additional ESs were reanalyzed using ESM. Because such an increase in the number of ESs over polar regions in winter was not possible with K_dPAR, we used another method (see below).

Step 2. To estimate PAR_{0+}, the algorithm first identified ESs (C_{np}) within a Target Zone (TZ). The boundary of TZ around a TES was done by calculating the distance between each ES and the TES in all environmental dimensions. Only ESs that fall within a certain threshold distance, represented by the vector $D_p = [d_i]$, were retained for the analysis. The threshold distance, which can vary along each dimension, must be between 0.01 and 0.5, but is often between 0.01 and 0.2, depending on the number of available ESs, the level of smoothing w chosen for estimations while keeping the local relationship among variables (Beaugrand, 2024).

Step 3. PARs were obtained from the ESs located within the TES as follows:

$$\hat{a} = \frac{\sum_{i=1}^{t} w_i a_i}{\sum_{i=1}^{t} w_i} \tag{9}$$

Where $\hat{a}$ was the estimated PAR_{0+}, t the total number of ESs within the TZ of a TES, i the ith ES within the distance imposed byD_p and w_i was the ith weight associated to a_i. w_i were assessed as follows:

$$w_i = ve^{-\lambda k_i} \tag{10}$$

With v being the maximum weight for a nil distance between a TES and an ES. The multivariate distance k between a TES and an ES was calculated as follows:

$$k_i = \sum_{j=1}^{p}\left|h_j - q_j\right| \tag{11}$$

With h ($0 \le h \le 1$) and q ($0 \le q \le 1$) being the values of each environmental parameter standardized between 0 and 1 between a TES (h) and an ES (q). The parameter λ was fixed so that $w_i = 1$ at $k_i = d_{max}$, with d_{max} assessed as follows:

$$d_{max} = \sum_{i=1}^{p} d_i \tag{12}$$

With p being the total number of environmental dimensions and d_i the threshold of distance fixed for each ecological dimension ($1 \le i \le p$). Using Eq. 13, λ was determined as follows:

$$\lambda = \frac{ln(v)}{d_{max}} \tag{13}$$

Therefore, the model depends upon the selection of two thresholds: (i) v the maximum weight for a distance $k_i = 0$; and (ii) threshold of distance $Dp = [d_i]$ with $0 < d_i \le 0.5$ for each ecological dimension i. Moreover, using this method also gives a maximum weight to additional ESs (*i.e.*, NASA OBPG individual sensor and MSDCS = 0) when

reanalyzed.

The choice of the best thresholds for d_i and v was made by comparing the statistical metrics computed on 10% of valid data originated from four observed days of each year of the period 1998–2022. (Supplementary Table 1). The four days were winter and summer solstices, and spring and fall equinoxes. All pairs of combinations for $d_i = [0.05; 0.1; 0.15; 0.2]$ and $v = 10^{\alpha}$, with α being ranged from 2 to 30 every step of 1, were tested; a value of d_i below 0.05 could lead to an absence of estimates and above 0.2 to overfitting.

We compared our ESM-based procedure with another model called the Generalized Regression Neural Network (GRNN) (Specht, 1991). GRNN is a powerful tool for nonlinear modeling, capturing complex relationships and patterns in the data (Specht, 1991). In the model, the architecture is determined by the number of input and output variables, with neuron patterns mirroring data patterns (Antanasijević et al., 2014). We selected the GRNN due to its established efficacy in robust water quality predictions (Antanasijević et al., 2014; Csábrági et al., 2017) and previous comparison with ESM in plankton modeling (Beaugrand, 2024). However, it is sometimes considered as a black box model (Koutsoyiannis, 2007), with potential for overfitting depending on parameter selection (Cao and Zhang, 2021; Wu et al., 2015). We applied the GRNN on winter and summer solstices, and spring and fall equinoxes using thirteen values of smoothing (*i.e.*, spread): spread = [0.0001; 0.001; 0.005; 0.01; 0.05; 0.1; 0.2; 0.3; 0.5; 0.7; 1; 2; 3] (Supplementary Table 2).

A visual comparison was carried out between estimated missing data and the original Globcolour products to evaluate if abrupt changes or inconsistencies at the edges of data outlines were observed (Fig. 1). At the end of this procedure, we had a PAR_{0+} product (1998–2022) of 721 Latitudes × 1440 Longitudes × 1Depth × 9122 Days (365 days × 25 years) with no missing data.

3.3.2. PAR_{0+} comparison with independent data

Methodology was evaluated by first comparing PAR_{0+} estimations (Globcolour, NASA OBPG and ESM) with *in-situ* data from the COVE platform (Supplementary Fig. 1) (Rutledge et al., 2006). Then, we utilized the EPIC DSCOVR PAR_{0+} data specifically for the period 2015–2022 (Supplementary Fig. 2). For both analyses, a special attention was made on differentiating the ESM estimations based on the availability of additional NASA OBPG PAR_{0+} data to gain insights on the performance of the ESM models and the impact of incorporating additional NASA OBPG PAR_{0+} data.

3.3.3. Generation of a gap-free K_dPAR product

To estimate missing K_dPAR data, we employed an alternative approach based on (Globcolour) Chla concentration. Remote sensing of Chla concentration typically lacks coverage in high latitudes during winter, resulting in missing Chla estimates within the ESM due to the absence of ESs in these zones and no supplementary ESs could be incorporated, akin to those used for PAR_{0+}. Therefore, we applied Data Interpolating Empirical Orthogonal Functions (DINEOF, Alvera-Azcárate et al., 2005) for interpolating and extrapolating the Chla daily data for 1998–2022.

DINEOF employs Empirical Orthogonal Functions (EOFs) derived from Singular Value Decomposition (SVD) to represent data. It reconstructs data using weighted EOFs, minimizing residuals (Beckers and Rixen, 2003). DINEOF handles large multivariate datasets and effectively interpolates/extrapolates complex spatial-temporal data (Alvera-Azcárate et al., 2007). DINEOF was applied to 25 years (1998–2022) of sliding matrices centered on the 60 days of Chla data precededing and following it. Initially, missing data were replaced by 0, which introduced a significant bias into our data. This bias was particularly pronounced in high latitudes during winter, leading to the underestimation of Chla concentration. Consequently, we chose to substitute missing data by modelled Chla from the Mercator-Ocean biogeochemical hindcast model (Beckers and Rixen, 2003). A comparison was performed, focusing specifically on valid Chla data above a SZA threshold of 65° (Supplementary Fig. 3). The initial hindcast Chla model values were also included in the analysis. An additional visual comparison was carried out between the original Globcolour product, the hindcast Chla model and both DINEOF methods used to estimate missing Chla data (Fig. 2). Finally, K_dPAR was computed by estimating K_d at 490 nm (K_d490, m^{-1}) from Chla concentration, and finally, from K_d490 (Morel et al., 2007):

$$K_d490 = 0.0166 + 0.077298 \times Chla^{0.67155} \quad (14)$$

$$K_dPAR = 0.0665 + \left((0.874 \times K_d490) - {}^{0.00121}/_{K_d490} \right) \quad (15)$$

Morel's equations were chosen for estimating Chla, prioritizing Case 1 waters. However, they may inadequately represent Case 2 waters due to difficulties in capturing light attenuation variability. Although challenging for global daily retrieval pre-remote sensing era, approaches based on remote sensing reflectance inversion may offer better results for coastal waters (Singh et al., 2022).

Completing missing PAR_{0+} Globcolour data by using averaged OBPG mono-sensor data and interpolating the remaining gaps with DINEOF gave performance metrics that were less satisfactory compared to the ESM (Supplementary Table 3). To carry out the DINEOF on PAR_{0+} data, we replaced missing data either by zeros (RMSD = 28.09; MAD = 21.53) or MERRA-2 PAR_{0+} data (RMSD = 27.98; MAD = 21.43), which were above the RMSD/MAD values for the ESM (RMSD = 20.30; MAD = 13.82). We remind that for Chla data, we used DINEOF because additional OBPG data were lacking, which explains why using DINEOF provided more accurate estimates than ESM.

At the end of this procedure, we had a surface K_dPAR product (1998–2022) of 721 Latitudes × 1440 Longitudes × 1 Depth of 9122 Days (365 days × 25 years) with no missing data.

3.3.4. K_dPAR comparison with independent data

To ensure validity of high latitudes interpolated Chla data, an additional analysis was conducted based on Chla originating from BGC-Argo data (Supplementary Fig. 4). This analysis firstly computed statistical metrics between original Chla data from Globcolour L4 product and BGC-Argo Chla data. Then, similar statistical metrics were computed for BGC-Argo data with no corresponding Globcolour data, *i.e.* interpolated using DINEOF.

3.4. Period including the portion of time previous to remote sensing (1958–2022, daily resolution)

3.4.1. PAR and K_dPAR estimations since 1958

For each geographical cell with sea ice concentration (SICONC) < 0.1 (thereafter called the target cell) and for each target day of a specific year (1958–2022), we estimated the PAR_{0+} and K_dPAR by considering a geographical square of 31 cells (*i.e.* 15 cells on the western/eastern and northern/southern sides of the target cell). The target cell was characterized by the following combination of environmental variables (TES): MSDCS, MCC, and TCL for PAR_{0+} and BTM, DTC, WST, and SST for K_dPAR (see Supplementary Table 8 for the meaning of acronyms).

For a target day and year (1958–2022), PAR_{0+} and K_dPAR were estimated from ESs within a geographical square for this specific day, considering the time period 1998–2022 as a reference. Therefore, ESs belonged to a 3D matrix of 31 geographical squares (south-north) x 31 geographical squares (west-east) x 25 reference days in 1998–2022. Note however that the number of reference days was 24 when the considered year belonged to 1998–2022, to exclude ESs of that year to ensure that TESs and ESs were independent. PAR_{0+} and K_dPAR in geographical cells with sea ice concentration (SICONC) ≥ 0.1 were not estimated.

The choice of the best thresholds to use for d_i and v was made by comparing the statistical metrics on outputs for winter and summer

solstices, and spring and fall equinoxes of the period 1998–2022 (Supplementary Tables 4 and 6). All pairs of combinations for $d_i = [0.05; 0.1; 0.15; 0.2]$ and $v = 10^{\alpha}$, where α ranged from 2 and 30 every step of 1, were tested, but d_i was set to 0 for BTM and DTC to filter ESs by class in the computation of K_dPAR. Then, the comparison was performed between the ESM and the GRNN for PAR_{0+} and K_dPAR, utilizing identical training sets (Supplementary Tables 5 and 7). The visual comparison with the original Globcolour products was renewed to help determine the level of agreement/discrepancy between the derived spatial patterns obtained with ESM and remote sensing products (Figs. 3–4).

Finally, four indicators available with ESM computation were applied to evaluate ESM uncertainty: (i) The total number of ES around each TES (*i.e.* within the target zone), a higher number of ES implying a more reliable estimation; (ii) the weighted standard deviation and (iii) the weighted coefficient of variation of estimates, and (iv) the average multivariate distance between ES and TES; a small average distance suggesting that estimations are likely to be more accurate. Detailed descriptions are available in Beaugrand (revised) (Supplementary Fig. 5–7).

At the end of the procedure, we obtained a PAR_{0+} and K_dPAR product (1958–2022) of 721 Latitudes × 1440 Longitudes × 23,741 Days (*i.e.*, 365 days × 64 years) with a few missing data, which were completed using the inverse squared distance interpolation (Wackernagel, 2003).

3.4.2. Test and bias correction

Statistical analyses were performed to test the accuracy and correct any bias present in the ESM surface outputs for 1958–2022. This process involved multiple steps, including comparisons with *in-situ* data, existing products, and EOF analyses to understand the spatiotemporal patterns of the estimations.

3.4.2.1. Comparison with in-situ data, bias correction, and alignment with accepted models (daily 1958–2022).

ESM outputs were compared to *in-situ* data from the COVE platform (Rutledge et al., 2006) (Supplementary Fig. 8). A linear correction was subsequently applied to ESM-based PAR_{0+} to improve its reliability. The corrected ESM-based PAR_{0+} product was compared to MERRA-2 reanalysis data (Supplementary Fig. 9).

After an additional comparison with the original uncompleted Globcolour dataset (Supplementary Fig. 10) K_dPAR outputs from our ESM-based modelled were subsequently confronted to *in-situ* data. To comparison purposes, these outputs were transformed into K_d490 values as only *in-situ* K_d490 data were available from SEABASS. The resulting association facilitated a comprehensive comparison with corresponding values from Globcolour (Supplementary Fig. 11).

ERA5 A_s was originally computed for the total solar spectral domain from SSR and SSRD (Eq. 2). We then estimated MERRA-2 surface Albedo in the PAR spectral domain (A_{sv}) from MERRA-2 A_s by means of a linear regression analysis (Supplementary Fig. 12). The resulting slope and intercept coefficients were subsequently utilized to obtain ERA5 A_{sv} from ERA5 A_s, which allowed us to assess PAR_{0-} as follows:

$$PAR_{0-} = PAR_{0+} \times (1 - A_{sv}) \quad (16)$$

Finally, Z_{eu} was computed using PAR_{0-} and K_dPAR assessed from our ESM-based models. Z_{eu} was the depth at which PAR_z (*i.e.*, PAR at a given depth z) equals 1% of PAR_{0-}. PAR_z was computed using the Beer-Lambert law for the exponential extinction of light in water (Ingle and Crouch, 1988), assuming a constant K_dPAR along the water column:

$$ln PAR_z = ln PAR_{0-} - K_d PAR \times Z \quad (17)$$

ESM Z_{eu} were subsequently matched with Z_{eu} *in-situ* data from SEABASS (Fig. 5). The comparison was conducted to both the raw products of Globcolour and the biomass content hindcast model from Mercator-Ocean, which used surface Chla concentration as input to estimate Z_{eu} (Morel and Berthon, 1989).

3.4.2.2. EOF on estimated surface PAR (monthly 1958–2022).

An EOF analysis was conducted on a 20-years (1997–2017) monthly time series of seasonal and non-seasonal (anomalies) PAR_{0+} over the Tropical Oceans, without rotation of the data (Tan and Frouin, 2019; Supplementary Figs. 13–14). Results were associated with El Niño indices and compared with previous findings (Tan and Frouin, 2019). Another EOF examined the 1982–1983 super El Niño's impact on PAR_{0+} in the tropical Pacific (Chertock et al., 1991) (Fig. 6).

3.5. Estimation of PAR at depth

Once PAR_{0-} and K_dPAR were available for 1958–2022, PAR was computed along depth every 5 m from the surface up to 250 m by means of the Beer-Lambert law (Eq. 17) following methodology used by Gattuso et al. (2006, 2020) and presented in Fig. 7. At the end of the procedure, we obtained a final PAR product (1958–2022) of 721 Latitudes × 1440 Longitudes × 51 Depths × 23,741 Days (365 days × 64 years) with no missing data.

4. Results

4.1. Estimation of missing data in the Globcolour surface PAR dataset (1998–2022)

4.1.1. Influence of the distance to target string and maximum weight and comparisons between ESM and GRNN

We first examined how the distance to the target string d_i and the maximum weight v affected the estimates of missing PAR_{0+} data. After testing various combinations, we found that the best results were achieved when d_i was 0.05 and maximum v was 10^{30} (a value of $v > 10^{30}$ sometimes gave missing data, Supplementary Table 1). The performance metrics using this combination showed good results, with PRMSD = 18.58%, PMAD = 12.74%, MB = 0.052 W.m^{-2} and $r^2 = 0.83$.

The best statistical metrics for GRNN were found with a spread of 0.001 (Supplementary Table 2). While GRNN gave slightly better PMAD (12.20%) and MB (0.047 W.m^{-2}) results, the ESM outperformed for other metrics such as PRMSD (20.63%) and r^2 (0.80). Overall, CI was lower for ESM (CI = 0.16) compared to GRNN (CI = 0.18). Moreover, the computation time required for GRNN was much longer (~12 min) than with ESM (< 3 min). All r^2 were significant (*i.e.*, $p < 0.001$).

4.1.2. Visual agreement with data

A visual inspection between the completed dataset from the ESM ($d_i = 0.05$ and $v = 10^{30}$) and Globcolour for the four days (Spring and Autumn Equinoxes, Winter and Summer Solstices) of 2018 gave satisfactory results (Fig. 1). Specifically, at mid-latitudes, PAR_{0+} variability caused by clouds was observed, even when data were missing due to sensor geometry. The ESM estimated low PAR_{0+} values under cloudy conditions, while a maximum PAR_{0+} was reached in the absence of atmospheric obstacles. These data agreed well with observed PAR_{0+} data from EPIC DSCOVR. Moreover, at high latitudes, the ESM outputs displayed excellent continuity with both the Globcolour product and EPIC DSCOVR.

4.1.3. Comparisons with independent data

Original uncompleted Globcolour PAR_{0+} (1998–2022) were compared with *in-situ* data from the COVE platform (Supplementary Fig. 1a). The comparison revealed that the raw Globcolour dataset (1998–2022) performed well despite a loss in spatial resolution, with $r^2 = 0.91$, PMAD = 10.46% and PRMSD = 15.24%. A substantial MB of 10.27 W.m^{-2} was found, however, indicating a systematic bias. Data originating from OBPG (mean between MODIS and SeaWiFS) showed

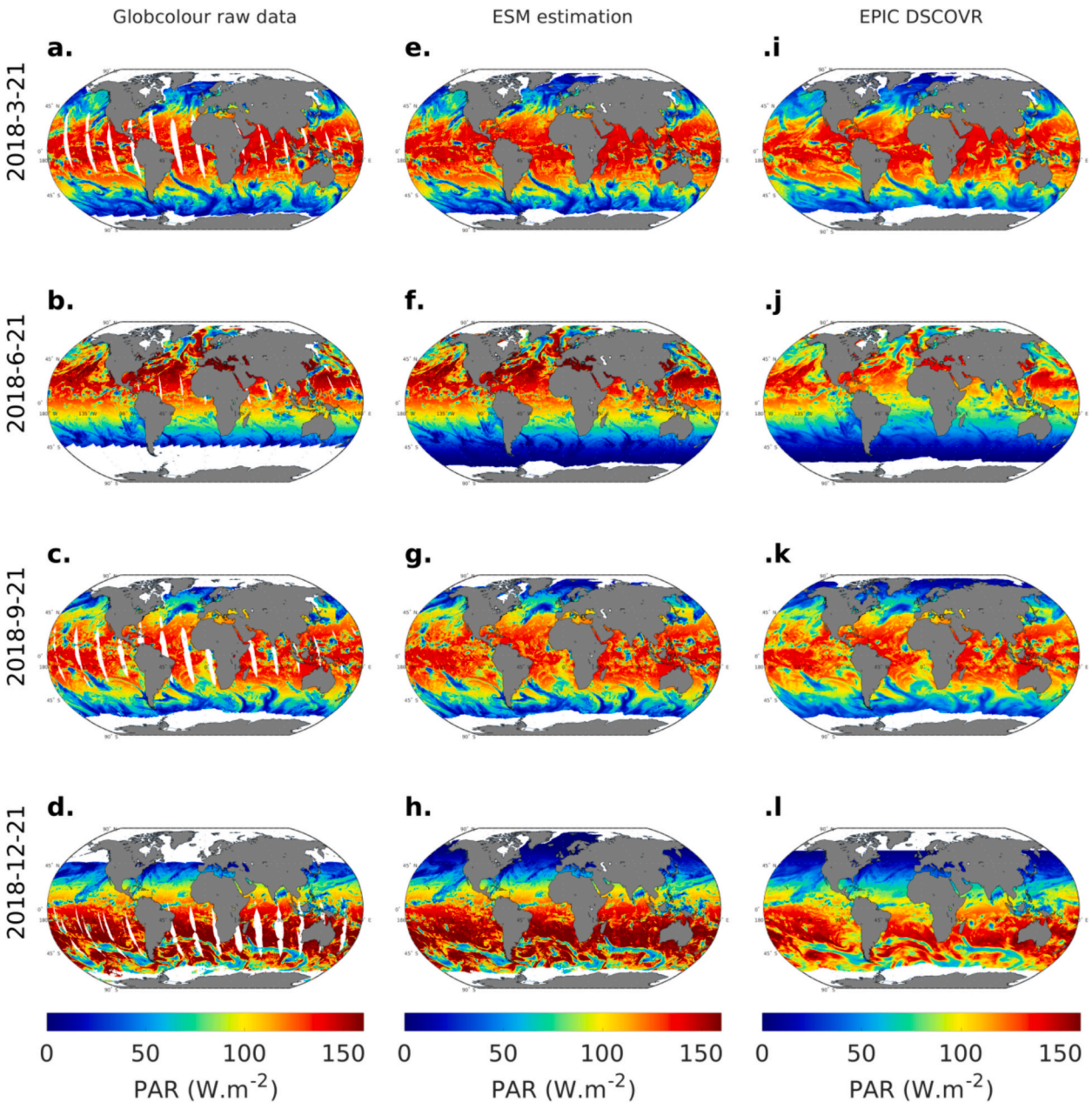


Fig. 1. *Global spatial distribution of PAR_{0+} ($W.m^{-2}$) for four days in 2018* (Spring and Autumn Equinoxes, Winter and Summer Solstices) observed by Globcolour (a-d), completed with the ESM (e-h) based on best parameters fitting (distance to target string $d_i = 0.05$ and maximum weight $v = 10^{30}$) and observed by EPIC DSCOVR (i-l). Data located in areas with SICONC $\geq$0.1 were removed.

slightly lower statistical metrics, with $r^2 = 0.9$, PMAD = 12.55% and PRMSD = 17.03% (Supplementary Fig. 1b). ESM-based PAR_{0+} showed good agreement with *in-situ* data, whether OBPG data were available (PMAD = 10.77%, PRMSD of 15.35%; Supplementary Fig. 1c) or not (PMAD = 11.28%, PRMSD of 16.28%; Supplementary Fig. 1d). All r^2 were significant (*i.e.*, $p < 0.001$).

Additional analyses were carried out to compare our results with PAR_{0+} originating from EPIC DSCOVR for 2015–2022 (Supplementary Fig. 2). A first analysis was made to compare the Globcolour raw product and the EPIC DSCOVR product. Despite a RMSD (PRMSD) of 16.52 W.m^{-2} (18.52%) and a MAD (PMAD) of 11.84 W.m^{-2} (13.28%), a strong r^2 of 0.85 was found between the two datasets, with the majority of data points close to the identity line (Supplementary Fig. 2a). Regarding the comparison with *in-situ* data, comparison with OBPG data presented slightly lower statistical metrics, with a r^2 of 0.84, a MAD (PMAD) of 12.64 W.m^{-2} (14.17%) and a RMSD (PRMSD) of 17.4 W.m^{-2} (19.51%) (Supplementary Fig. 2b). The EPIC DSCOVR product was then compared with our ESM estimations. When we did not have additional OBPG data, comparing EPIC DSCOVR and ESM estimations gave high positive correlation ($r^2 = 0.81$), a MAD (PMAD) of 12.29 W.m^{-2} (13.78%) and a RMSD (PRMSD) of 17.3 W.m^{-2} (19.4%) (Supplementary Fig. 2c). When NASA OBPG data were used to complete ESs, comparison showed strong results with $r^2 = 0.85$, MAD (PMAD) = 11.2 W.m^{-2} (12.56%) and RMSD (PRMSD) = 15.74 W.m^{-2} (17.66%) (Supplementary Fig. 2d). All r^2 values were highly significant ($p < 0.001$).

4.2. Estimations of missing data in the Globcolour K_dPAR dataset (1998–2022)

We estimated missing K_dPAR by applying DINEOF to Chla data. To quantify the influence of using modelled Chla data instead of replacing missing data by 0 in the initial stage of DINEOF, the specific approach

described in Section 3.3.3 was applied (Supplementary Fig. 3). The findings revealed that using DINEOF by replacing missing values by the Mercator-Ocean hindcast model Chla data led to improved estimations for the valid data points above the SZA threshold (SZA $\geq$65°) of almost 5% in PMAD and 8% in PRMSD. Directly replacing missing data by the Mercator-Ocean hindcast model values yielded suboptimal results.

A visual inspection also confirmed that Chla values were better estimated for high latitudes when DINEOF replaced initial missing data by the Mercator-Ocean hindcast model values instead of 0 (Fig. 2b, f *versus* Fig. 2d, h).

An additional comparison was conducted between Chla originating from BGC-Argo data, with a distinction made between original data from Globcolour and interpolated data (*i.e.* where no Globcolour data were available) using DINEOF with our methodology (Supplementary Fig. 4). Statistical metrics revealed comparable performance for both DINEOF interpolated (RMSD = 0.44, MAD = 0.34 and $r^2 = 0.41$) and Globcolour Chla data (RMSD = 0.43, MAD = 0.36 and $r^2 = 0.48$), highlighting the consistency between completed Chla values with DINEOF and the Globcolour Chla product.

K_dPAR was subsequently estimated using Morel's equations (Eqs. 14

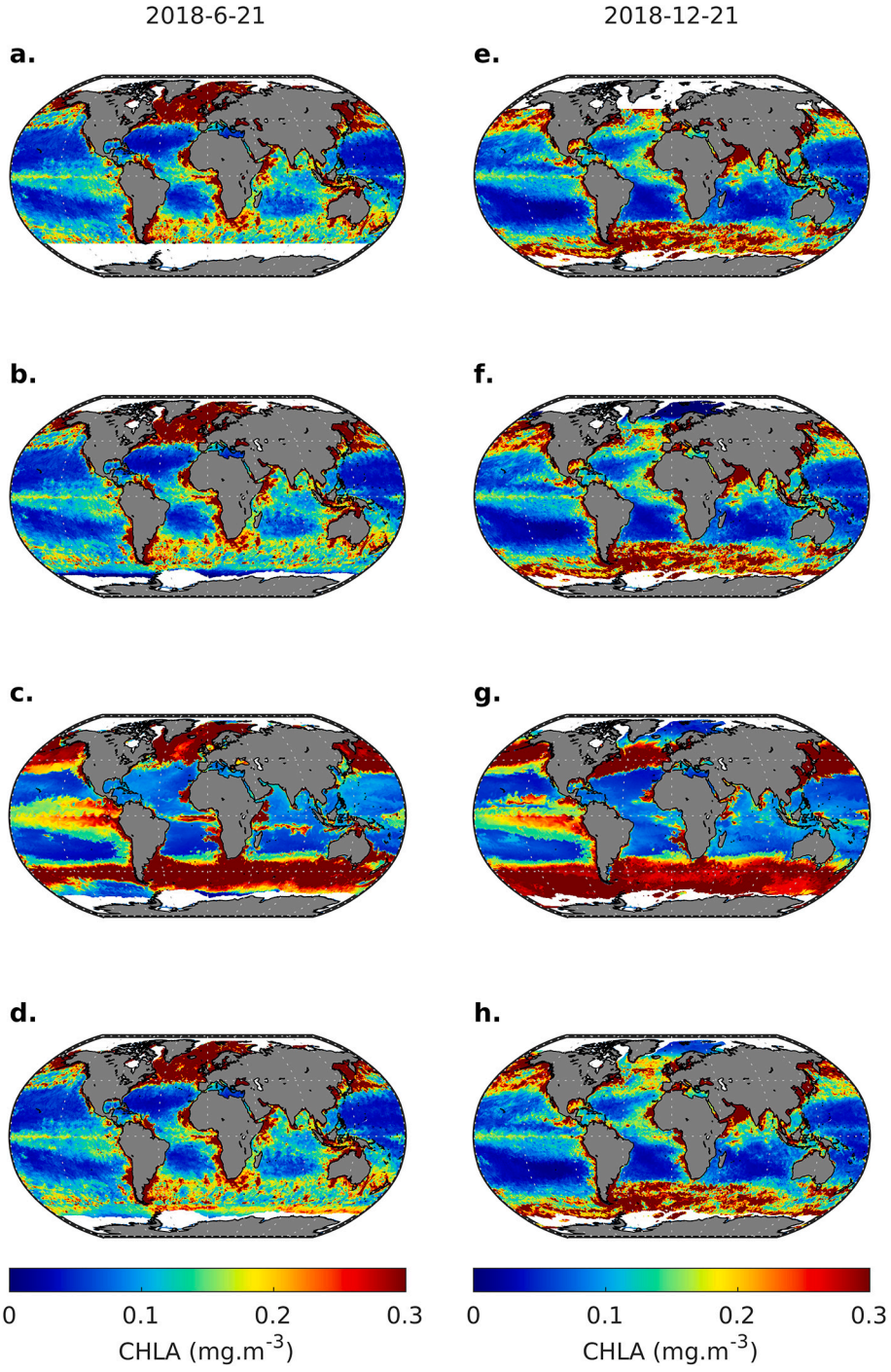


Fig. 2. Global spatial distribution of Chla (mg.m^{-3}) for two days (Summer and Winter Solstices) of 2018 observed from Globcolour (a, e), and completed using DINEOF with missing data initially replaced by 0 (b, f), Chla from the Mercator-Ocean hindcast model (c, g), and completed using DINEOF with missing data initially replaced by Chla from the Mercator-Ocean hindcast model (d, h). Data located in areas with SICONC $\geq$0.1 were removed. Chla color bar was set with a threshold at 0.3 mg.m^{-3} (~18% data saturated).

and 15, 2007) on the Chla product.

4.3. Modelled surface PAR and K_dPAR for 1958–2022

4.3.1. Influence of the distance to target string and maximum weight and comparisons between ESM and GRNN

4.3.1.1. PAR. We first examined the influence of the distance to target string d_i for each environmental node and maximum weight v, (Eqs. 12–13; Supplementary Table 4). Although the best statistical metrics were found with d_i=1, we chose to set the maximum d_i to 0.2 to keep local relationship among variables and avoid overfitting. Therefore, CI was lowest for $d_i = 0.2$ and $v = 10^2$ for PAR_{0+} estimations, with PMAD = 20.03%, PRMSD = 26.23%, and r^2 indicating that 66% of the variance was explained by the model. The best spread for GRNN was 2 for PAR_{0+}, with PMAD = 19.94%, PRMSD = 24.03%, and r^2 indicating that 67% of the variance was explained by the model. (Supplementary Table 5). GRNN gave similar performance metrics in comparison with the ESM, with $d_i = 1$ and $v = 10^2$, except for the longer computation time required for GRNN. A systematic MB with an underestimation of the outputs was found for both methods. All correlation coefficients were significant ($p < 0.001$).

4.3.1.2. K_dPAR. The influence of the distance to target string d_i for each environmental node and maximum weight v were also conducted for K_dPAR using SST, WST, BTM and DTC as explanatory variables (Supplementary Table 6). Ultimately, a value of $d_i = 0.15$ and $v = 10^5$ was chosen for the ESM because it was associated to the best CI, with PMAD = 10.92%, PRMSD = 21.59%, and $r^2 = 0.74$. Although there were only minor differences among the two methods, the ESM generally performed slightly better than GRNN, which gave the best results for a spread of 0.05, with PMAD = 11.08%, PRMSD = 22.22%, and $r^2 = 0.73$ (Supplementary Table 7). All correlation coefficients were significant ($p < 0.001$).

It is worth noting that there were still some missing data in the modelled PAR_{0+} and K_dPAR datasets for 1958–2022 (<1< 1% per day). These missing data were ultimately estimated using inverse squared distance interpolation.

4.3.2. Visual comparisons with observations in 2018

4.3.2.1. PAR. A visual comparison between Globcolour and our ESM-based PAR_{0+} revealed a clear smoothing of PAR_{0+} by the ESM (Fig. 3). Overall, our ESM-based dataset (1958–2022) reproduced well the latitudinal and seasonal changes in PAR_{0+}. PAR_{0+} variability did not seem to be reached in extreme atmospheric conditions, particularly in high latitudes during summer or within the Intertropical Convergence Zone (ITCZ) where low PAR_{0+} were expected. However, the ITCZ

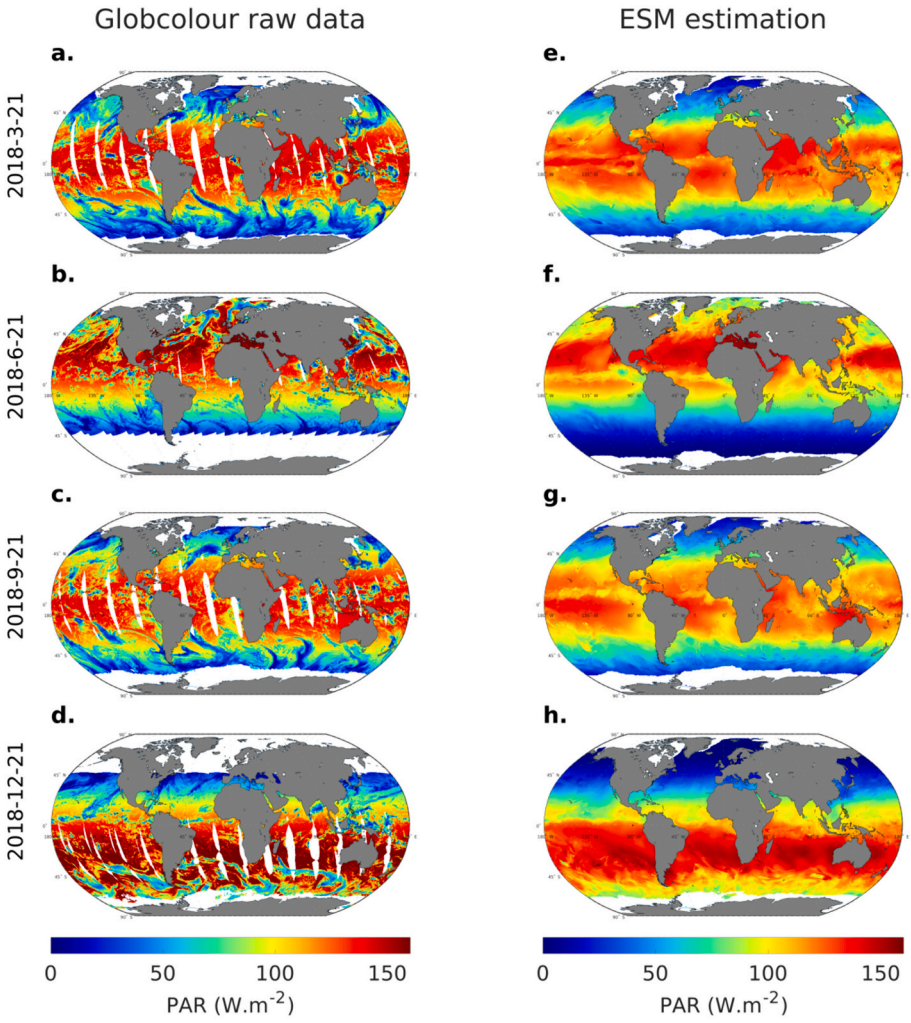


Fig. 3. Global spatial distribution of PAR_{0+} ($W.m^{-2}$) for four days of 2018 (Spring and Autumn Equinoxes, Winter and Summer Solstices) observed by Globcolour (a-d) and modelled using our ESM (e-h) based on best parameters fitting (distance to target string $d_i = 0.2$ and maximum weight $v = 10^2$). Data located in areas with SICONC ≥ 0.1 were removed.

remained detectable with PAR_{0+} of ~50 $W.m^{-2}$, especially in June and September (Fig. 3 f,g). A visual animation of PAR_{0+} ($W.m^{-2}$) evolution over the year 2018 is available in Supplementary Videos (Supplementary Video 1).

The reliability of PAR estimates was also evaluated with four key indicators provided by ESM (Supplementary Fig. 5). The first indicator provided the number of ES contributing to estimates in each geographical area (Supplementary Fig. 5a-d). Throughout the four days of 2018, the number of ES was generally high globally, with exceptions in specific regions characterized by strong ERA5 cloud cover, particularly near the equator. The second indicator examined the weighted standard deviation of abundance estimates (Supplementary Fig. 5e-h). Since higher PAR values correlated with greater weighted standard deviation (Supplementary Fig. 5e-h *versus* Fig. 3e-h), we looked at the weighted coefficient of variation (Supplementary Fig. 5i-l). This latter indicator was higher at high latitudes and near the equator. Latitudinal disparitites were also noted in the assessment of mean latitudinal PRMSD and PMAD between Globcolour PAR_{0+} data and ESM based model PAR_{0+}, with errors particularly high polewards (see Supplementary Fig. 6). Lastly, the mean distance to the target string revealed that, typically, the mean distance fell within the range of 20–40% (Supplementary Fig. 5 m-p). However, this indicator was notably lower over tropical regions (below 10%).

4.3.2.2. K_dPAR. The visual comparisons between observed and spatial patterns of ESM-based K_dPAR supported results provided by performance metrics (Supplementary Table 6). Similar geographical patterns and seasonal variabilities were observed among the two products (Fig. 4). K_dPAR values seemed to reach their maximum along coasts. ESM-based K_dPAR data reproduced well the plume observed in equatorial regions (Fig. 4). Moreover, seasonal changes were well reconstructed with significant light attenuation occurring during winter at high latitudes and inversely. Finally, low K_dPAR allowed a clear identification of the main oceanic gyres, while high K_dPAR were observed in upwellings. A visual animation of surface K_dPAR (m^{-1}) evolution over the year 2018 is available in Supplementary Videos (Supplementary Video 2).

The four key indicators were also computed for K_dPAR (Supplementary Fig. 7). The number of ES contributing to estimates was generally high, except in coastal regions influenced by BTM and DTC filters (Supplementary Fig. 7a-d). The weighted coefficient of variation (Supplementary Fig. 7i-l) was greater in coastal areas, over Northern Latitudes in June and over Southern Latitudes in December. Last, mean distance to target string (Supplementary Fig. 7 m-p) typically ranged from 30% to 50% globally and correlated with WST anomalies.

4.3.3. Test and bias correction

We also performed three more additional analyses to evaluate the quality of our PAR_{0+} and K_dPAR datasets.

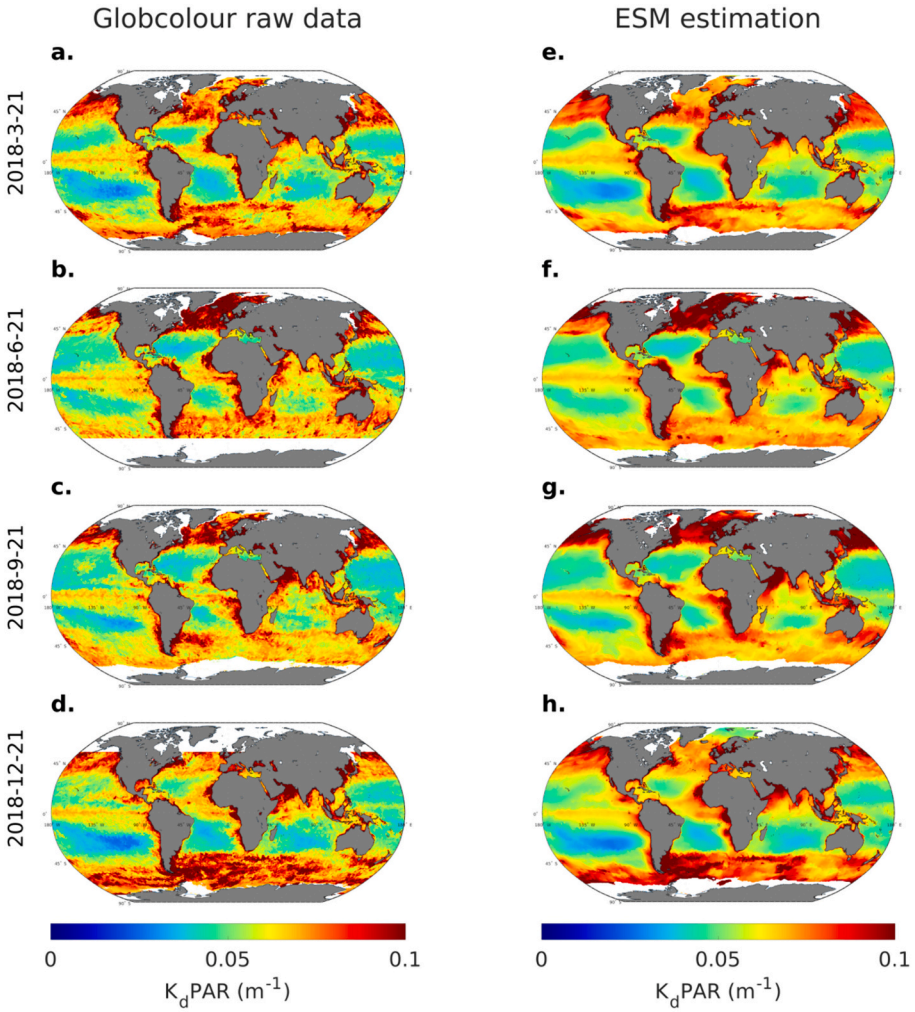


Fig. 4. Global spatial distribution of K_dPAR (m^{-1}) for four days of 2018 (Spring and Autumn Equinoxes, Winter and Summer Solstices) observed by Globcolour (a-d) and modelled using our ESM (e-h) based on best parameters fitting (distance to target string $d_i = 0.15$ and maximum weight $v = 10^5$). Data located in areas with SICONC $\geq$ 0.1 were removed. K_dPAR color bar was set with a threshold at 0.1 m^{-1} (<6% data saturated).

4.3.3.1. Comparison with in-situ data (daily 1958–2022)

4.3.3.1.1. PAR. The comparison between ESM-based modelled data (1958–2022) and *in-situ* measurements at the COVE site from 2005 to 2008 revealed a significant bias (*i.e.*, underestimation of the measurements), despite a high positive correlation (see Supplementary Fig. 8a in contrast with Fig. 1d). We corrected this bias by means of a linear model calculated between our ESM-based PAR_{0+} and COVE *in-situ* data. The computed slope from the linear model was 1.10 and a revised comparison was subsequently performed (Supplementary Fig. 8b). The correction improved the ESM-based PAR_{0+} data and the values of PMAD and PRMSD diminished by ~4%. Most importantly, the mean bias substantially decreased to -2.61 W.m^{-2}. We therefore applied the correction to the whole ESM-based PAR_{0+} dataset for 1958–2022.

Further analysis involved comparing PAR_{0+} data from MERRA-2 and our ESM-based dataset (1958–2022) with the original uncompleted Globcolour dataset (1998–2022; Supplementary Fig. 9). PAR_{0+} assessed from MERRA-2 and our ESM-based PAR_{0+} dataset displayed a high degree of similarity in their global statistical metrics when compared to Globcolour, with CI = 0.26 and CI = 0.28, respectively. Our ESM-based PAR_{0+} dataset showed a slight improvement over MERRA-2 concerning PMAD but was less efficient when comparing PRMSD and r^2. Our ESM-based dataset globally overestimated Globcolour PAR_{0+} with a MB of 9.39 W.m^{-2} whereas MERRA-2 globally underestimated Globcolour PAR_{0+} with a MB of -9.52 W.m^{-2}. The density plot also revealed a close alignment of the majority of data points with the identity line, suggesting a better correspondence for PAR_{0+} originated from our ESM-based dataset, except for extremely low PAR_{0+} values.

4.3.3.1.2. K_d490-K_dPAR. While the additional density plot comparing K_dPAR from our ESM-based dataset with the original uncompleted Globcolour dataset confirmed the good correspondence between ESM and Globcolour K_dPAR ($r^2 = 0.81$, Supplementary Fig. 10), no K_dPAR data were available in the SEABASS validation database for *in-situ* comparison. We chose to use available K_d490 data to indirectly compare our modelled K_dPAR dataset (1958–2022), using Morel's model (Eq. 15, 2007) to compute K_d490 values from the ESM and the original uncompleted Globcolour (1998–2022) datasets. A total of 910 matchups were obtained and examined in Supplementary Fig. 11; note that the data were transformed using a decimal logarithm prior to statistical comparison. The statistical metrics showed a good agreement between the *in-situ* data and the ESM-based dataset. The performance of the Globcolour raw dataset was slightly better, with a CI = 0.19 instead of CI = 0.22 for the ESM-based dataset. Noteworthily, there was no significant constant bias observed in the ESM-based dataset, although it slightly underestimated high K_dPAR values, as suggested by MB = -0.05 m^{-1}.

4.3.3.1.3. Z_{eu}. Validation of Z_{eu} was also important to ensure the accuracy of our ESM-based PAR and K_dPAR datasets (1958–2022). MERRA-2 albedo was initially employed to evaluate ERA5 surface albedo within the visible spectrum. The coefficients determined through a linear model relating MERRA-2 A_s and A_{sv} demonstrated favorable statistical metrics, elucidating approximately 60% of the variance, with a PRMSD of 10.17% and a PMAD of 7.98% (Supplementary Fig. 12). This approach was found to be sufficient for our study and subsequently the coefficients derived from this linear model were applied to ERA5 surface albedo data. Then, Eq. 16 was applied to calculate PAR_{0-} and Eq. 17 was utilized to compute PAR at depth, ultimately determining the value of Z_{eu}. An inter comparison between *in-situ* SEABASS data and estimations of Z_{eu} from our ESM-based datasets, the raw Globcolour datasets, and Z_{eu} estimated from the Mercator-Ocean hindcast model was conducted (Fig. 5). A total of 403 valid matchups was found for the four datasets.

The comparisons of Z_{eu} from our ESM-based datasets, raw Globcolour datasets, Mercator-Ocean hindcast model, and the *in-situ* SEABASS dataset showed relatively satisfactory PRMSD, PMAD and r^2 values. Despite the metrics computed for Globcolour Z_{eu} exhibiting the best statistical metrics, a marginal discrepancy of approximately 2 m between the Globcolour outputs and Z_{eu} estimated from the Mercator-Ocean hindcast model and ESM was identified although the Mercator-Ocean hindcast model encountered challenges in the precise retrieval of high Z_{eu} estimates.

4.3.3.2. Comparisons of our datasets through existing studies (monthly 1958–2022). We used our ESM-based PAR_{0+} dataset (1958–2022) to investigate long-term changes (seasonal and non-seasonal) in PAR_{0+} in the Tropical Pacific by means of the methodology described in Tan and Frouin (2019) and Chertock et al. (1991), who performed EOFs on satellite and solar radiation (radiative transfer model) data, respectively. Similar temporal changes in PAR_{0+} (*i.e.*, ESM-based modelled PAR_{0+} *versus* satellite PAR_{0+} and ESM-based modelled PAR_{0+} *versus* modelled solar radiation) suggest that our dataset is reliable at these scales.

We first performed an EOF analysis to examine long-term seasonal changes in PAR_{0+}. Similar to Tan and Frouin (2019), results, showed latitudinal disparities mainly induced by SZA (Supplementary Fig. 13). EOF modes highlighted positive anomalies in Asia and ~ 10°N in the East Atlantic and Pacific (Supplementary Fig. 13c-d), and negative loadings in North Australia and West-South America (Supplementary Fig. 13e-f).

We also examined long-term non-seasonal changes in PAR_{0+} using an EOF (Supplementary Fig. 14). Our findings agreed with the

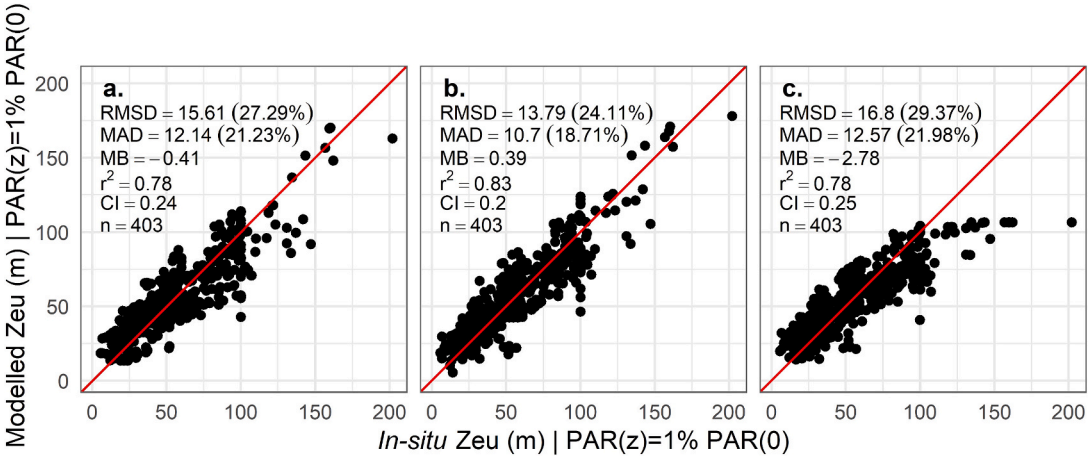


Fig. 5. Comparisons between in-situ euphotic depth (Z_{eu}) and (a) Z_{eu} calculated from PAR_{0-} and K_dPAR datasets modelled with ESM (1958–2022), (b) Z_{eu} calculated from the Globcolour PAR_{0-} and K_dPAR original incomplete datasets and (c) Z_{eu} from the Mercator-Ocean hindcast model. Points are matchups selected from the SEABASS global validation dataset (1998–2007).

correlations reported in Tan and Frouin (2019). The first EOF mode spatial pattern (accounting for 23.93% of the total variance, Supplementary Fig. 14a) was marked by a dipolar pattern, exhibiting negative loadings in the central tropical Pacific and positive loadings in the western tropical Pacific. Moreover, the amplitude of the first EOF mode identified the two major El Niño events of 1997–1998 and 2014–2015 and to a lesser extent the 2009–2010 El Niño event in the Tropical Pacific (Supplementary Fig. 14b). Long-term changes in PAR_{0+} were significantly correlated positively with the MEI index ($r_{Pearson} = 0.74$, $p < 0.001$ *versus* 0.84 in Tan and Frouin, 2019). The second EOF mode spatial pattern (accounting for 7.88% of the total variance, Supplementary Fig. 14c) unveiled positive loadings concentrated in the Central-eastern tropical Pacific region. The amplitude of the second EOF mode demonstrated a robust correlation with EMI (Supplementary Fig. 14d, $r_{Pearson} = 0.51$, $p < 0.001$ *versus* 0.48 in Tan and Frouin, 2019).

A second EOF was carried out on 1979–1985 to examine if our ESM-based PAR_{0+} identified the influence of the super El Niño event of 1982–1983 on PAR_{0+}, as per the study of Chertock et al. (1991). Most of the non-seasonal variability in PAR_{0+} was explained by the first two EOF modes. Our analysis was consistent with Chertock et al. (1991), suggesting that our ESM-based dataset (1958–2022) effectively retrieved non-seasonal variability, at least for this region and time period. Negative anomalies were observed over the central and eastern tropical Pacific Ocean, positive anomalies were observed in the western tropical Pacific and the peak of negative amplitude of the first mode was detected in February 1983 (Fig. 6). The correlation between non-seasonal EOF mode 1 and the SOI was significantly positive ($r_{Pearson} = 0.76$, $p < 0.001$).

4.4. Estimation of PAR at depth

Once PAR_{0+} was converted into PAR_{0-}, PAR was estimated at depth by means of Eq. 17, assuming a uniform K_dPAR across the water column. To visually represent the latitudinal daily PAR variation, four days were selected (Spring and Autumn Equinoxes, Winter and Summer Solstices) in 2018 along longitude of 27°W (Fig. 7). The transects captured well the latitudinal variation in PAR, with low PAR from the surface to 250 m over high latitudes during winter and high PAR from surface to 250 m towards mid-latitudes over subtropical the Atlantic Gyres. Low spread in PAR was observed over the equatorial zone due to high K_dPARs. The influence of the smoothing on our ESM-based dataset was clear. Differences in PAR between winter and summer solstices were also well identified. Light spread to greater depths during the Winter Solstice in the Southern Hemisphere with a maximum depth of 200 m. in the Northern Hemisphere, the threshold of 0.1 $W.m^{-2}$ reached a maximum depth of 160 m during the Summer Solstice. These differences were observed because of the varying values of both PAR and K_dPAR over this transect, with a less significant concentration of K_dPAR in the South Atlantic Gyre compared to the North. This led to deeper PAR propagation during the Winter Solstice in the Southern Hemisphere compared to the Summer Solstice in the Northern Hemisphere (Supplementary Fig. 15). A visual animation of PAR at depth ($W.m^{-2}$) evolution over the year 2018 along longitude of 27°W is available in Supplementary Videos (Supplementary Video 3).

5. Discussion

The aim of the study was to provide a comprehensive, gap-free 4D (Latitudes × Longitudes × Depth × Day) dataset of PAR for two distinct

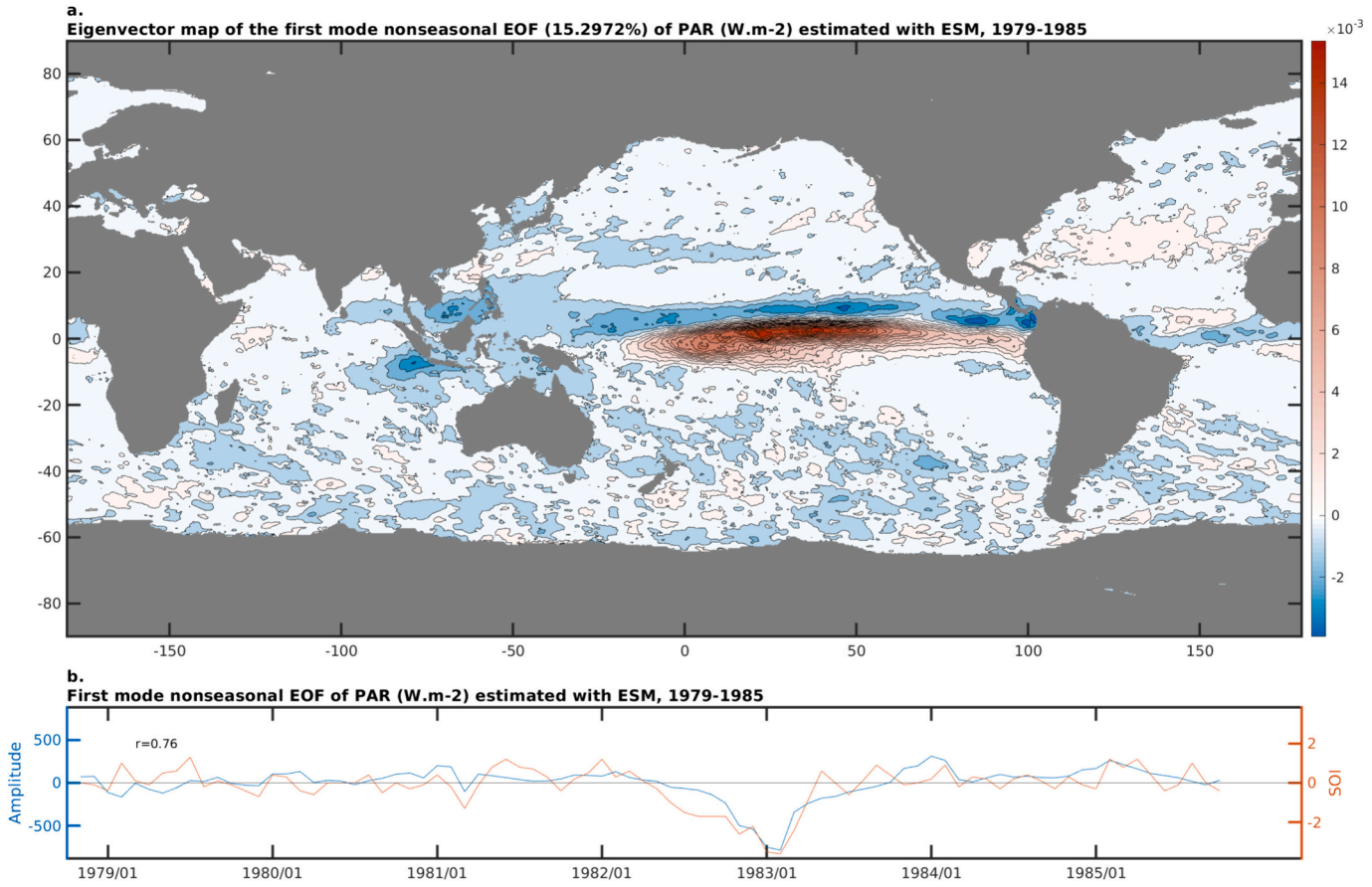


Fig. 6. Results of an EOF performed on non-seasonal PAR_{0+} (1979–1985) with (a) eigenvector map of the first EOF mode and (b) first EOF mode amplitude time series in relation with the SOI index.

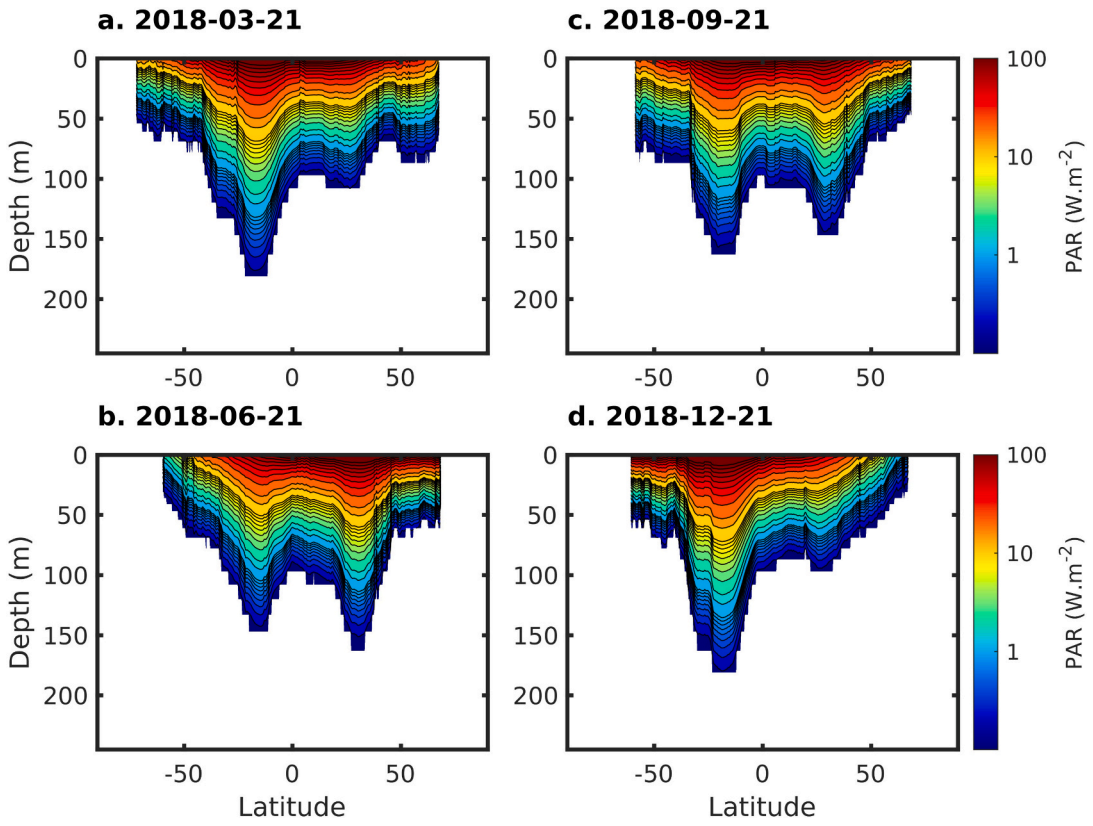


Fig. 7. PAR (W.m^{-2}) as a function of depth using PAR_{0-} and K_dPAR (m^{-1}) predicted from our procedures for four days of 2018 (Spring and Autumn Equinoxes, Winter and Summer Solstices) at longitude 27°W. A threshold of 0.1 W.m^{-2} was chosen for minimum light intensity. PAR values are expressed in $\log_{10}$ scale.

time frames: 1998–2022 (derived from completed satellite observations) and 1958–2022 (generated from modeling). The significance of such datasets is foundational in marine biology, as PAR plays a crucial role in evaluating the spatial and temporal variations of plankton biodiversity. The significance is especially pronounced in high latitudes regions where light availability can constrain plankton production at the surface (Caracciolo et al., 2021; Kléparski et al., 2022; Helaouet et al., 2016). Additionally, in the realm of climate change biology, PAR assumes critical importance as species may adapt to temperature shifts by altering their vertical distribution (Jónasdóttir and Koski, 2011). However, such adaptations are contingent upon PAR levels, which not only influences photosynthesis, but also dictates the vertical distribution of phytoplankton, a factor that intricately regulates the vertical distribution of zooplankton as well (Hintz et al., 2022; Sailley et al., 2015).

5.1. Our procedures

5.1.1. PAR

The ESM, originally devised for ecological niche modeling (Beaugrand, 2024), is a straightforward method suitable for interpolating and extrapolating in geographical space provided that combinations of environmental variables associated to PAR_{0+} or K_dPAR values (*i.e.*, an environmental string) exist. The two crucial parameters of the model are the distance to the target environmental string TES (d_i) and the maximum weight (v). In this study we explored numerous combinations fine tune and optimize the results.

We compared our ESM model with GRNN, a widely recognized tool renowned for its capacity to model complex data relationships. Both the ESM and GRNN methods were employed to fill missing data for Globcolour PAR_{0+}, with supplementary OBPG PAR_{0+} data sourced from individual sensors to increase the number of ESs in high latitude areas (Fig. 1). The merging of remote sensing data has already been explored through various techniques, such as weighted averaging (Beck et al., 2017) or machine learning (Wu et al., 2020), which have proven effective in minimizing disparities and biases in the individual monosensor datasets involved (IOCCG, 2007; Kwiatkowska and Fargion, 2003; Pottier et al., 2006). In the case of ESM, a low value of d_i (*i.e.* 0.05) combined to a maximum v of 10^{30} were essential for integrating additional OBPG data and accurately capturing PAR_{0+} variations at high latitudes (Fig. 1; Supplementary Table 1, Supplementary Fig. 1). The ESM-based modelled PAR_{0+} data for the period 1958–2022 were derived from a 25-year (1998–2022) dataset cube. Upon comparison between the ERA5 dataset and the remote sensing dataset, it became apparent that non-linear correlations existed, resulting in significant disparities in PAR_{0+} values for similar environmental conditions (ESs). However, Cawkwell and Bamber, 2002 provided evidence that simulated net radiation was also sensitive to subtle changes in cloud cover, while Wu et al. (2023) found strong correlations between ERA5 cloud cover and MODIS-A cloud cover, albeit based on annual averages. To address this issue, experiments were conducted to adjust the parameter d_i in the ESM, ultimately selecting a larger value of $d_i = 0.2$. This adjustment led to noticeable improvements in our estimates (Supplementary Table 4). The increased smoothing, also observed using the GRNN model with a spread of 3 (Supplementary Table 5), decreased PAR_{0+} accuracy under extreme atmospheric conditions, as depicted in the ITCZ (Fig. 3). This discrepancy becomes particularly evident when compared to MERRA-2 reanalysis data (Supplementary Fig. 9): although the analysis excluded low PAR_{0+} values in high latitudes during winter, the reliability of our ESM-based model in assessing low PAR_{0+} values associated with cloud conditions diminished when compared to the original Globcolour dataset. The omission of aerosols in our environmental parameters for PAR_{0+} estimations, due to their unavailability in the ERA5 dataset, may potentially lead to discrepancies for the period spanning 1958 to 2022 when comparing our results to other products, as atmospheric aerosols play a crucial role in light transmittance, particularly under clear sky, affecting both the direct and diffuse components

of solar irradiance (Foyo-Moreno et al., 2023; Twomey, 1974; Albrecht, 1989; Lohmann and Feichter, 2005). Lastly, remote sensing data from high latitudes are not as effectively processed by algorithms of atmospheric correction compared with those from low latitudes in winter.(Li et al., 2017; He et al., 2018). This resulted in similar uncertainties at high latitudes with higher PRMSD and PMAD values in these regions (Supplementary Figs. 5–7). The four indicators provided with the ESM may be used as quality flags on this uncertainty, allowing users the option to retain estimates derived from a specific number of ESs or those associated with a low weighted coefficient of variation. However, the four indicators provided with the ESM offer valuable insights that can serve as quality flags for the user, who has the option to retain estimates derived from a specific number of ESs or those associated with a low coefficient of variation.

5.1.2. K_dPAR

DINEOF could have been utilized to fill PAR_{0+} data from Globcolour; however, its reliance on limited EOFs posed a risk of small-scale data loss, particularly within large-area gaps. ESM, which integrates ERA5 and OBPG data, accounts for local relationships to provide accurate predictions and should be prioritized, especially when ancillary variables are available. Nevertheless, such variables were not accessible for K_dPAR estimations due to various factors such as the SZA threshold, cloud cover, and atmospheric errors (Liu and Wang, 2022). Consequently, we resorted to using DINEOF, initializing it differently by incorporating modelled Chla, enabling us to estimate values in high latitude regions. While estimations in these regions primarily relied on modelled Chla, comparative analysis demonstrated alignment with the original Globcolour product (Supplementary Figs. 3–4). It is important to acknowledge that the conversion algorithm proposed by Morel et al. (2007) is primarily suitable for open ocean waters, where K_dPAR values typically fall below 0.5 m^{-1}. However, for turbid coastal waters, the algorithm may not yield accurate results (Wang et al., 2009), highlighting a limitation that users should consider when using our datasets.

In contrast to PAR, we did not observe a consistent mean bias for K_dPAR spanning from 1958 to 2022 (Supplementary Table 6). This disparity arises from our utilization of DTC and BTM classes to estimate K_dPAR values in coastal regions. Our approach solely incorporated K_dPAR values from these coastal regions to compute the weighted mean, excluding lower K_dPAR values typically found in the open ocean. Moreover, employing a spatial window of 31 × 31 grid cells over a period of 25 years, enabled the model to directly utilize ESs within specific coastal water regions. This capability is crucial as these regions encapsulate a substantial portion of the optical variability observed in natural waters (Morel and Maritorena, 2001). Our analysis underscored the model's adeptness in accurately reproducing the main patterns observed in biogeographic studies (Longhurst et al., 1995; Longhurst, 1998) and effectively mimicking the primary characteristics of global Chla seasonal cycles (Fig. 4). Notably, our model exhibited pronounced light attenuation during the spring season, attributable to favorable environmental conditions in SST and an enriched mixed layer depth abundant in nutrients (following an increase in SST anomalies) contributing to the development of phytoplankton blooms (Martinez et al., 2011; Rumyantseva et al., 2015; Yoder et al., 1993). Blooms significantly influence the estimation of PAR at depth, resulting in higher K_dPAR values, a pattern that is well observed by satellite sensors (Gohin et al., 2015). Although our model successfully identifies the timing of these blooms in the open ocean, additional research is needed to evaluate their potential influence on light attenuation.

5.2. Test of our datasets

To further enhance the accuracy and reliability of our datasets, we conducted comparisons with independent remote sensing data available provided by the EPIC DSCOVR mission. However, distinguishing between uncertainties caused by the model or the data itself proved challenging at times. Comparisons with Globcolour raw data revealed clear disparities with EPIC DSCVOR data, primarily due to its global averaging from multiple observations during the day (Supplementary Fig. 2a). Nonetheless, these discrepancies remained globally consistent when assessing missing data estimations using our ESM, regardless of the availability of additional NASA OBPG data (Supplementary Fig. 2c-d). Given the advantages offered by EPIC's global coverage and continuous observations over traditional geosynchronous satellites for remote sensing in high-latitude regions (Yang et al., 2022), it warrants consideration for future PAR_{0+} estimations.

The comparison between the ESM-based modelled and *in-situ* PAR_{0+} data from COVE revealed an underestimation of PAR (Supplementary Fig. 8a), whereas Globcolour MB depicted PAR_{0+} overestimation (Supplementary Fig. 1a), mainly due to a biased water-leaving reflectance spectrum (Sun et al., 2010). To rectify this, we applied a linear regression model to correct our ESM-based modelled PAR_{0+} dataset for 1958–2022. No intercept was computed to avoid overestimation when PAR_{0+} values were close to zero (*i.e.* no solar light at high latitudes in winter). This correction significantly improved our product in terms of evaluation metrics (Supplementary Fig. 8b). Moreover, ESM-based modelled PAR_{0+} demonstrated satisfactory evaluation metrics when compared to the MERRA-2 PAR_{0+} product (Supplementary Fig. 9), serving as an important pre-requisite to assessing the product's robustness prior to the availability of remote sensing data.

To assess how our ESM-based modelled PAR_{0+} dataset (1958–2022) captured temporal variations (both seasonal and non-seasonal) in the spatial distribution of PAR_{0+}, we performed EOF analysis on monthly averaged PAR_{0+} (Fig. 6; Supplementary Fig. 13–14). Our findings were initially compared to related analyses conducted on some remote sensing products spanning 1997 to 2017 (Tan and Frouin, 2019), revealing similar results. This suggested that our dataset effectively reproduced the seasonal cycles in PAR_{0+}. The spatial pattern of PAR_{0+} exhibited strong seasonality, reflecting the influence of the Sun's position, which is a primary determinant of PAR_{0+} (Tan and Frouin, 2019).

Temporal changes in PAR_{0+} varied latitudinally (Supplementary Fig. 13). Generally, higher tropical latitudes displayed greater seasonal variability in PAR_{0+} compared to lower latitudes, with PAR_{0+} reaching its peak levels during summer (Supplementary Fig. 13, Tan and Frouin, 2019). The seasonal EOF analysis highlighted these patterns, while the non-seasonal EOF identified the relation between canonical and Modoki El Niño events with PAR_{0+} patterns (Supplementary Fig. 14). The significant positive correlations between MEI and the first mode of non-seasonal EOF (Supplementary Fig. 14b) suggested that our PAR_{0+} dataset reconstructed well the influence of canonical El Niño events characterized by increased high cloudiness over the central and eastern tropical Pacific due to reduced oceanic upwelling and elevated sea surface temperatures, which promote atmospheric convection (Li et al., 2016; McPhaden et al., 2011). Furthermore, the substantial positive correlations between EMI and the second mode of non-seasonal EOF (Supplementary Fig. 14d) are consistent with alterations in high cloud cover and SST anomalies observed during Modoki El Niño events, (Li et al., 2016; Ashok et al., 2007).

While the merged data from Globcolour, utilized as inputs in the ESM-based datasets, remain uncertain for trend analysis (ACRI-ST, 2007), the positive correlations with EMI and MEI indices, in addition to the successful identification of prominent El Niño events by our model (Fig. 6), offer an initial assessment of the dataset ability to detect climate signals despite the challenges in estimating daily PAR_{0+} variability. This provides a robust foundation for its application in plankton modeling across extensive temporal scales. Indeed, capturing the interannual variation of the ENSO phenomenon is crucial, as it may result in a 30–40% decrease in primary productivity, as observed in the Gulf of California during the super 1997–1998 El Niño (Kahru et al., 2004). Accurately representing these significant hydro-meteorological events is particularly relevant for a comprehensive understanding of the dynamics and responses of plankton populations to ENSO-driven

variations in environmental conditions. However, further long-term analyses are still necessary to determine the suitability of ESM-based datasets for climate investigations because inter-mission differences, studied in Chla trends based on merged (OC-CCI) and mono-sensor time series (*e.g.*, SeaWiFS Chla; Mélin et al., 2017), along with artifacts found in GlobColour bias-correction (Maritorena et al., 2010; Hammond et al., 2018), may introduce artificial patterns in time series (Mélin, 2016).

Our ESM-based modelled 4D PAR dataset (1958–2022) enabled us to estimate Z_{eu}, a crucial parameter in biogeography and plankton ecology due to its influence on primary production (Behrenfeld and Falkowski, 1997; Platt and Sathyendranath, 1988). Our estimates of Z_{eu} closely aligned to those from the Mercator-Ocean hindcast model (Fig. 5) which utilized SST and PAR from the ERA-INTERIM reanalysis dataset (CMEMS, 2022) and computed Z_{eu} from Chla using Morel's case I model (Morel and Berthon, 1989). The similarity in Z_{eu} estimations between our ESM-based modelled 4D PAR dataset and the Mercator-Ocean hindcast model suggests that our approach captured well the light availability at depth. In our approach, we assumed that K_dPAR was constant throughout the water column. However, estimating Z_{eu} accurately requires accounting for the variability of K_dPAR with depth. Similar challenges arise when utilizing Morel and Berthon's eq. (1989) based on surface Chla to compute Z_{eu} from remote sensing surface Chla (Morel and Maritorena, 2001; Lee et al., 2007). Despite this, the Mercator-Ocean hindcast model Z_{eu} model underperformed for high Z_{eu} estimations, particularly in open-ocean conditions where water clarity is optimal. In contrast, our ESM and Globcolour estimations demonstrated better performance (Fig. 5). The implementation of the Beer-Lambert Law in our case was deemed more suitable, as it was commonly utilized for open-ocean conditions, while its applications in coastal waters may introduce larger discrepancies in Z_{eu} estimation (Lee et al., 2007).

6. Conclusion

Our study presents two comprehensive 4D PAR (gap-free) datasets from 1998 to 2022 and 1958 to 2022, featuring daily temporal resolution and 0.25° latitude x 0.25° longitude spatial resolution, encompassing depths up to 250 m. The PAR dataset covering the 1998–2022 period exhibits more precision compared to its 1958–2022 counterpart, thus warranting its preference for studies focusing on the period post-1998. Despite the relatively reduced daily variability of our 1958–2022 PAR dataset in contrast to the 1998–2022 dataset, it adeptly reconstructs several patterns observed in previous studies and exhibits significant correlation with modelled and *in-situ* data. These 4D PAR (gap-free) datasets offer considerable potential for the reconstruction of plankton spatial distribution, particularly in high latitude regions, where previous PAR data limitations hindered accurate modeling of plankton distribution and potential responses to global climate change. Leveraging a global depth PAR distribution may facilitate (i) deeper insights into key phytoplankton dynamic processes, including the depth limitations of photosynthetic organisms, which remains relatively unexplored and (ii) the examination of variations in the light field within sensitive ecosystems.

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CRediT authorship contribution statement

Jérôme Castant: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Vincent Vantrepotte:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition. **Robert Frouin:** Writing – review & editing, Writing – original draft, Validation. **Grégory Beaugrand:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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